

Drivers in the selection of the Heating System for the California households

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1 Introduction

I work in a consulting firm in San Diego (California), more precisely in the statistical analysis department, and a client has asked for a report about some important drivers for his business. In my firm that's the core business.

This report is conducted in 2003 for a client firm working in the field of the installation of residential heating systems and is currently facing a rapidly evolving market landscape. In the last years market in this sector is changing rapidly and, in order to increase the profits and decrease the unproductive investments, the company needs to understand the underlying mechanisms that guide consumer preferences when choosing a heating technology. So, the central matter of this report is a discrete choice problem, where we have finite number of choices and we know the choice specific characteristics and some other characteristics that change between the choice makers.

My colleagues who work in the data collection department have created a survey because we didn't have any data on the heating systems. They have distributed the survey to 1,000 households in California, sent only to local families, because our client firm works only in California. After eliminating the missing values and the aberrant values, we have obtained a sample with clean data of 900 households. The respondent must choose between five distinct technological alternatives: Gas Central (GC), Gas Room (GR), Electric Central (EC), Electric Room (ER), and the increasingly relevant Heat Pump (HP) systems. These technologies are the 5 principal options offered by our client firm.

Ultimately, this report aims to provide the client with a predictive framework to forecast market shares under different pricing scenarios and to design targeted marketing strategies that align with the diverse needs of modern homeowners.

2 Methodology

2.1 Data Collection

As we said, the data set is composed of 900 records of choices made by each household, stored in the column “depvar”. Then the alternative-specific attributes are the installation cost (“IC”) and the annual operating cost (“OC”). Furthermore, the dataset includes socio-economic and geographic indicators for the households: Annual household income divided in class, the age of the head of the household, the number of rooms in the dwelling and the specific region of residence (north coastal, south coastal, mountain, or valley). These variables allow us to account for the "heterogeneity" of the choice makers, acknowledging that a high-income family in a coastal area might prioritize different features compared to a household in a colder, mountainous region. Moreover, for the sake of precision, each household has been assigned to a level of income dividing the income by 10,000 and then assigned to the nearest round number.

The data set (2.1) has been modified in order to make the modeling and the interpretation easier, so every socio-demographic variable has been transformed into a dummy variable. For “region” has been created a dummy variable for every region, instead for “income”, “age” and “room” the levels of each variable have been grouped in two groups and then transformed in a dummy. “income” becomes a dummy variable “high_income” = 1 if “income” is 5 or higher, “agehed” becomes a dummy variable “old” = 1 if “agehed” is 50 or higher and “rooms” becomes a dummy variable “many_rooms” = 1 if “rooms” is 5 or higher.

Table 2.1: Heating system Database

Statistic	N	Mean	St. Dev.	Min	Max
idcase	900	450.500	259.952	1	900
ic.gc	900	776.827	115.563	431.830	1,158.900
ic.gr	900	921.770	138.087	574.940	1,344.000
ic.ec	900	824.543	125.274	469.610	1,230.500
ic.er	900	983.928	147.155	546.820	1,496.300
ic.hp	900	1,046.481	156.703	532.320	1,679.800
oc.gc	900	172.116	25.940	84.016	248.430
oc.gr	900	154.471	22.886	77.863	227.920
oc.ec	900	476.803	73.153	237.440	705.360
oc.er	900	429.730	65.793	179.950	664.430
oc.hp	900	219.299	32.970	120.970	318.580
income	900	4.641	1.685	2	7
agehed	900	42.944	14.095	20	65
rooms	900	4.424	1.744	2	7
regionmountn	900	0.113	0.317	0	1
regionmcostl	900	0.289	0.453	0	1
regionscostl	900	0.401	0.490	0	1
regionvalley	900	0.197	0.398	0	1
high_income	900	0.540	0.499	0	1
low_income	900	0.460	0.499	0	1
old	900	0.409	0.492	0	1
young	900	1.000	0.000	0	1
many_rooms	900	0.478	0.500	0	1
few_rooms	900	0.522	0.500	0	1

2.2 Model Specification

From a methodological standpoint, the principal model used for the analysis has been the Multinomial Logit (MNL) model, the most well-known and used model for discrete choice modeling. We obviously know that the MNL model has some strict assumptions, such as, for example, the independence of errors across choice alternatives, which derives from the i.i.d. EV1 distribution assumption of the errors; in social science environments, most of the time it is difficult to test these assumptions. But at the same time, the MNL model is very parsimonious, it has a closed-form solution for the maximum likelihood, and is highly interpretable—all desirable characteristics.

To increase the precision of our MNL, it is crucial to specify the components of the deterministic utility for each choice in the best possible way. So, to have the best specification, I have tried five different MNL models, but for the sake of brevity I will highlight only the most important.

The first one was a model where I included all the independent variables of our dataset; in the dataset, only two are choice-dependent variables, while the others are socio-demographic variables. My decision to use all the possible predictors for the dataset is based on the fact that in economic theory, all variables could be significant in the personal utility. For example, I hypothesized that a higher age of the head of the household could lead to a stickiness in the decision to switch from a less innovative heating system, such as Gas Central, to a more innovative one, such as Electric Central. But this was not the case: all socio-demographic variables were not significant except for one; the income level was significant in the utility of the Gas Room (“GR”) choice.

Instead, the installation and operating costs were significant for every choice utility. In the next model, I kept both costs for every choice and I get rid of all the socio-demographic variables except for the income level in the utility specification of the “GR” choice. Socio-demographics can be added to the utility formula on their own, but this leads to a difficult interpretation of the results; so, it is common sense to apply them as a “shift” of another attribute to make a better interpretation of the results possible.

So, after testing some possible interaction specifications for the variable “high_income,” the best model was the “MNL single interaction,” where the interaction is with the operating cost for the gas room (“OC”), but only for the utility of GR.

$$\begin{aligned}
U_{n,GC} &= \alpha_{GC} + \beta_{IC} \cdot IC_{n,GC} + \beta_{OC} \cdot OC_{n,GC} \\
U_{n,GR} &= \alpha_{GR} + \beta_{IC} \cdot IC_{n,GR} + (\beta_{OC} + \beta_{OC,Income} \cdot High_Income_n) \cdot OC_{n,GR} \\
U_{n,EC} &= \alpha_{EC} + \beta_{IC} \cdot IC_{n,EC} + \beta_{OC} \cdot OC_{n,EC} \\
U_{n,ER} &= \alpha_{ER} + \beta_{IC} \cdot IC_{n,ER} + \beta_{OC} \cdot OC_{n,ER} \\
U_{n,HP} &= 0 \quad + \beta_{IC} \cdot IC_{n,HP} + \beta_{OC} \cdot OC_{n,HP}
\end{aligned}$$

3 Results and discussion

I also considered the possibility of a more parsimonious model, excluding the interaction between the “oc” and “high_income” in the gas room utility function.

$$\begin{aligned}
U_{n,GC} &= \alpha_{GC} + \beta_{IC} \cdot IC_{n,GC} + \beta_{OC} \cdot OC_{n,GC} \\
U_{n,GR} &= \alpha_{GR} + \beta_{IC} \cdot IC_{n,GR} + \beta_{OC} \cdot OC_{n,GR} \\
U_{n,EC} &= \alpha_{EC} + \beta_{IC} \cdot IC_{n,EC} + \beta_{OC} \cdot OC_{n,EC} \\
U_{n,ER} &= \alpha_{ER} + \beta_{IC} \cdot IC_{n,ER} + \beta_{OC} \cdot OC_{n,ER} \\
U_{n,HP} &= 0 + \beta_{IC} \cdot IC_{n,HP} + \beta_{OC} \cdot OC_{n,HP}
\end{aligned}$$

After fitting this model, called “Micro MNL”, I have looked at the results from the LR test between the two models (Table 3.1); the slightly more complicated one seems better. Also, under indicators such as the AIC or BIC (which is very punitive for the model complexity), “MNL single interaction” seems the best.

Table 3.1: Likelihood Ratio (LR) Test Results

Model	Log-Likelihood (LL)	Parameters (K)	χ^2	df	p -value
Micro MNL	−1008.23	6	—	—	—
MNL Single Interaction	−1005.36	7	5.74	1	0.0166

*Significance at the 5% level

However, recognizing that consumer preferences are rarely uniform, I also consider the implementation of a Mixed Logit (Random Parameters Logit) model. This advanced approach allows us to capture unobserved heterogeneity and relax the restrictive "Independence of Irrelevant Alternatives" (IIA) assumption.

But after the estimation, as can be seen from Table 3.2, the sigma parameters of the ML model’s predictors are not significant. Consequently, we cannot conclude that the Betas exhibit inter-individual heterogeneity; therefore, in this case, the Mixed Logit would be meaningless to use. Moreover, the ML model relaxes some other MNL assumptions, such as the lack of correlation in unobserved utility over time. However, because the dataset in this case does not consist of panel data, this assumption is plausible.

Table 3.2: Mixed Logit Model Estimation Results

Parameter	Estimate	Rob. s.e.	Rob. <i>t</i> -ratio	<i>p</i> -value
<i>Random Parameters (Normal Distribution)</i>				
$\mu_{\beta_{IC}}$ (Mean Invest. Cost)	-0.001526**	0.000608	-2.510	0.012
$\sigma_{\beta_{IC}}$ (S.D. Invest. Cost)	-0.000093	0.000182	-0.509	0.611
$\mu_{\beta_{OC}}$ (Mean Oper. Cost)	-0.006964***	0.001466	-4.751	<0.001
$\sigma_{\beta_{OC}}$ (S.D. Oper. Cost)	0.000034	0.000117	0.290	0.772
<i>Alternative Specific Constants (ASC)</i>				
$ASC_{GasCentral}$	1.714180***	0.222142	7.717	<0.001
$ASC_{GasRoom}$	0.535458**	0.225000	2.380	0.017
$ASC_{ElectricCentral}$	1.653308***	0.439885	3.759	<0.001
$ASC_{ElectricRoom}$	1.847459***	0.348928	5.295	<0.001
$ASC_{HeatPump}$ (Reference)	0.000000	—	—	—
<i>Interactions</i>				
$\beta_{OC \times Income}$	-0.002927**	0.001243	-2.354	0.019
<i>Model Statistics</i>				
Observations	900			
Final Log-Likelihood (LL)	-1005.36			
ρ^2 (vs equal shares)	0.3059			
Adj. ρ^2 (vs equal shares)	0.2997			
AIC	2028.71			
BIC	2071.93			

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ (based on Robust *t*-ratios)

Therefore, even after these two attempts, the best model obtained from this study remains the “MNL single interaction.”

3.1 MNL results and interpretation

Let’s examine the estimation results from the “MNL single interaction”. Remember that the interpretation of the absolute magnitudes of all coefficients is influenced by the scale parameter, so it is not possible to interpret the absolute values of the coefficients β . It’s possible to interpret the sign and the marginal rate of substitution between attributes, which are in the logit models.

Table 3.3: MNL Single Interaction Model Estimation Results

Parameter	Estimate	Rob. s.e.	Rob. <i>t</i> -ratio	<i>p</i> -value
<i>Attribute Parameters</i>				
β_{IC} (Installation Cost)	-0.001525**	0.000608	-2.509	0.012
β_{OC} (Operative Cost)	-0.006964***	0.001466	-4.751	<0.001
<i>Alternative Specific Constants (ASC)</i>				
$ASC_{GasCentral}$	1.714197***	0.222139	7.717	<0.001
$ASC_{GasRoom}$	0.535415**	0.225020	2.379	0.017
$ASC_{ElectricCentral}$	1.653223***	0.439869	3.758	<0.001
$ASC_{ElectricRoom}$	1.847353***	0.348923	5.294	<0.001
$ASC_{HeatPump}$ (Reference)	0.000000	—	—	—
<i>Interactions</i>				
$\beta_{OC \times Income}$	-0.002927**	0.001243	-2.354	0.019
<i>Model Statistics</i>				
Observations	900			
Final Log-Likelihood (LL)	-1005.36			
ρ^2 (vs equal shares)	0.3059			
Adj. ρ^2 (vs equal shares)	0.3011			
AIC	2024.71			
BIC	2058.33			

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ (based on Robust *t*-ratios)

Starting from the ASCs, all the values estimated are positive and significant. This means that there is an intrinsic preference for the other alternative with respect to the Heating Pump. It is also plausible to say that some of the unexplained variance of our data has been captured by the ASCs. The estimate of the two β of the model represent a cost, so I was expecting that both of them would have been negative, that's consistent with the economic theory.

At first glance we can clearly see that the value for β_{OC} is bigger than the one of β_{IC} , but to understand the real magnitude of the estimated values of the installation and operative Cost, as I have previously said we need the Marginal Rate of Substitution (MRS).

$$MRS = \frac{\beta_{OC}}{\beta_{IC}} = \frac{-0.006964}{-0.001525} \approx 4.56$$

In this case the MRS is called Willingness to pay, because there is a cost at the denominator. We can say from the result that a person, on average, is willing to pay 4.56 dollars in installation cost for a decrease of operative cost of only 1 dollar.

Furthermore, is really interesting to look at the negative and significant interaction term ($\beta_{OC \times Income} = -0.002927$). This shows that the effect of operative costs is not uniform across the sample, the high income households become even more sensitive to operative costs, experiencing a steeper drop in

utility when recurring costs rise.

$$\frac{\beta_{OC \times Income}}{\beta_{OC}} = \frac{-0.002927}{-0.006964} \approx 0.42$$

The magnitude of the $\beta_{OC \times Income}$ is nearly 42% of the β_{OC} . This result runs contrary to classical economic theory, which suggests that high-income households should be less price-sensitive. We can attempt to explain this finding by considering that wealthier households typically have more capital readily available to change their heating systems. Consequently, they are more averse to increases in operative costs because they have a wider range of viable alternatives to choose from.

Here there are the Market shares for every alternative:

Table 3.4: Market Shares by Alternative

Alternative	Proportion	Percentage (%)
Gas Central (GC)	0.6367	63.67%
Gas Room (GR)	0.1433	14.33%
Electric Room (ER)	0.0933	9.33%
Electric Central (EC)	0.0711	7.11%
Heat Pump (HP)	0.0556	5.56%
Total	1.0000	100.00%

3.2 Own and Cross Elasticity

As anticipated, to evaluate the sensitivity of the probabilities to changes in the operational cost, we compute the direct elasticity. The generic formula for the own elasticity of the Operational Cost (OC) for an individual n and a generic alternative j is defined as follows:

$$\varepsilon_{n,j}^{\text{own}} = \beta_{OC} \cdot OC_{n,j} \cdot (1 - P_{n,j})$$

By applying this exact equation to our dataset, I calculated the point elasticity for every single observation. From these values, I then derived the mean of the own elasticity for each specific choice available in the model (Gas Central, Gas Room, Electric Central, Electric Room, and Heat Pump). Let's see the results:

Table 3.5: Mean Own Elasticity of Operational Cost by Alternative

Alternative	Mean Own Elasticity ($\bar{\epsilon}^{\text{own}}$)
Gas Central (GC)	-0.4326
Gas Room (GR)	-0.9222
Electric Central (EC)	-3.0961
Electric Room (ER)	-2.7261
Heat Pump (HP)	-1.4436

The own elasticity in the Logit models represent how much in percentage the initial probability of the decision maker to pick a specific choice change for an increase in one of the factor of 1%. In this case for example for the GC alternative, if the OC increase of 1% the probability of the decision maker to take that decision decrease of $0.6367 \cdot 0.4326 \cdot 0.01 \approx 0.0028$.

Following the analysis of direct effects, I calculate the cross elasticity to understand how a change in the operational cost of one specific alternative affects the probability of choosing any of the other available options. Also in this case the value of the elasticity represent a percentage change in the percentage to pick an alternative. Due to the IIA assumption, this effect is uniform across all competing alternatives. The generic formula for the cross elasticity with respect to the Operational Cost (OC) of an alternative j for an individual n is given by:

$$\epsilon_{n,j}^{\text{cross}} = -\beta_{OC} \cdot OC_{n,j} \cdot P_{n,j}$$

Then I obtain the mean cross elasticity for the operational cost of each alternative, as reported in Table 3.6.

Table 3.6: Mean Cross Elasticity of Operational Cost by Alternative

Alternative changing cost (j)	Mean Cross Elasticity ($\bar{\epsilon}^{\text{cross}}$)
Gas Central (GC)	0.7660
Gas Room (GR)	0.1535
Electric Central (EC)	0.2243
Electric Room (ER)	0.2665
Heat Pump (HP)	0.0836

So an increase of 1% in the operational cost for the Gas central, increase in percentage the probability of another alternative. For example, increase the probability of ER alternative by $0.0933 \cdot 0.7660 \cdot 0.01 \approx 0.0007$, so from 0.0933 to 0.0940.

3.3 Scenario simulation

Now I can study how the data would be affected by an increase of 1% in the Operational Cost for the most chosen alternative, Gas Central. That's officially

called scenario simulation.

Table 3.7: Comparison of Market Shares: Original vs. 1% Increase in GC Operational Cost

Alternative	Proportion		Percentage (%)	
	Original	New	Original	New
Gas Central (GC)	0.6367	0.6339	63.67%	63.39%
Gas Room (GR)	0.1433	0.1444	14.33%	14.44%
Electric Room (ER)	0.0933	0.0940	9.33%	9.40%
Electric Central (EC)	0.0711	0.0716	7.11%	7.16%
Heat Pump (HP)	0.0556	0.0560	5.56%	5.60%
Total	1.0000	1.0000	100.00%	100.00%

Now I can re-calculate the own elasticity for the Operational Cost for Gas Central, but this time thanks to this formula:

$$\varepsilon_{j,j'}^{\text{market}} = \frac{\ln(\sum_i P_{ij}^{\text{new}} / \sum_i P_{ij}^{\text{base}})}{\ln(1.01)}.$$

This formula uses the new market shares and the old market share to calculate the elasticities. This elasticity should coincide with the mean of individual elasticities, but it's different due to the nonlinearity of the MNL model. The same happens with the cross elasticity with respect to the Operational Cost for the Gas Central alternative, that is calculated with formula:

$$\varepsilon_{k,j'}^{\text{market}} = \frac{\ln(\sum_i P_{ik}^{\text{new}} / \sum_i P_{ik}^{\text{base}})}{\ln(1.01)}.$$

The scenario elasticities are the more precise ones because they rely on the actual simulation of the data, that's why they are the ones relevant for the market demand, and they are the ones we would use for policy or business analysis.

The results are showed here:

Table 3.8: Comparison of Elasticities: Mean Individual vs. Aggregate Market (1% GC Cost Increase)

Alternative	Elasticity w.r.t. Gas Central (GC) Cost	
	Mean Individual ($\bar{\epsilon}$)	Aggregate Market (ϵ^{market})
<i>Own Elasticity</i>		
Gas Central (GC)	-0.4326	-0.4320
<i>Cross Elasticity</i>		
Gas Room (GR)	0.7660	0.7667
Electric Central (EC)	0.7660	0.7355
Electric Room (ER)	0.7660	0.7371
Heat Pump (HP)	0.7660	0.7640

As expected from the IIA property of the MNL model, the individual-level cross elasticity is strictly uniform across all alternatives (0.7660). However, when calculating the new elasticity at the aggregate market level, slight discrepancies emerge (ranging from 0.7355 to 0.7667). This occurs because the aggregate formula accounts for the non-linear summation of individual probabilities before computing the logarithmic change, capturing the sample's heterogeneity more realistically. This also demonstrates that, as the Mixed Logit has already showed, that the IIA is a good approximation for our data, the differences in the cross elasticity are not strong, so the inter-individual heterogeneity is low for the operational Cost.

3.4 Elasticity stratification example

Next, I would like to show how elasticity could behave differently across different segments of the population because, as I previously clarified, this data frame contains many socio-demographic variables.

It would be interesting to see a quick example of a stratified study on direct elasticity. In this case, I am still considering the scenario of a 1% increase in the Gas Central Operational Cost, but this time I want to stratify the data by the age variable and compare the means of the two results.

As a stratification variable, I chose age because I think it is more probable that a younger person, compared to an older one, would have a more elastic demand.

Table 3.9: Aggregate Market Own Elasticity of Gas Central (GC) by Age Group

Age Group	Aggregate Market Own Elasticity (ϵ^{market})
Young	-0.4342
Old	-0.4289

It seems that I was right, the value for the older people is lower, so the

impact on the probability to stop using the Gas central system after an increase in the Operating Cost is slightly lower for the older people.

4 Conclusion

In conclusion, the MNL Single Interaction model provides strong and significant results for understanding heating system preferences among California households. The results confirm that installation and operative costs, together with the interaction between income and operative costs, are key factors in consumer choice. The model achieves a McFadden ρ^2 of 0.3059; in discrete choice modeling, a value higher than 0.25 is already considered a very good fit. Even with these positive results, future research could improve the analysis by adding new variables to the dataset. Including factors like the dwelling's insulation, past energy consumption, or environmental attitudes would help to capture more unobserved heterogeneity, creating an even better forecasting tool for the client.